

Problems on Dictionary

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to create a dictionary and display all key-value pairs.

Input	<code>{"name": "Amit", "age": 21}</code>
Expected Output	name: Amit age: 21

2. Write a Python program to add a new key-value pair to a dictionary.

Input	Dictionary = <code>{"name": "Amit"}</code> Add: <code>"city": "Kolhapur"</code>
Expected Output	<code>{"name": "Amit", "city": "Kolhapur"}</code>

3. Write a Python program to update the value of an existing key in a dictionary.

Input	Dictionary = <code>{"name": "Amit", "age": 20}</code> Update age = 21
Expected Output	<code>{"name": "Amit", "age": 21}</code>

4. Write a Python program to delete a key from a dictionary.

Input	Dictionary = <code>{"a": 1, "b": 2, "c": 3}</code> Delete key = b
Expected Output	<code>{"a": 1, "c": 3}</code>

5. Write a Python program to check whether a key exists in a dictionary.

Input	Dictionary = <code>{"name": "Amit", "age": 21}</code> Key = age
Expected Output	Key exists = True

6. Write a Python program to print all keys, all values, and all items of a dictionary.

Input	<code>{"a": 10, "b": 20}</code>
Expected Output	Keys = ['a', 'b'] Values = [10, 20] Items = [('a', 10), ('b', 20)]

7. Write a Python program to find the sum of all numeric values in a dictionary.

Input	<code>{"a": 10, "b": 20, "c": 30}</code>
Expected Output	Sum = 60

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8. Write a Python program to merge two dictionaries.

Input	D1 = {"a": 1, "b": 2} D2 = {"c": 3, "d": 4}
Expected Output	{"a": 1, "b": 2, "c": 3, "d": 4}

9. Write a Python program to count the frequency of characters in a string using a dictionary.

Input	aabcc
Expected Output	{'a': 2, 'b': 1, 'c': 2}

10. Write a Python program to create a dictionary from two lists: one of keys and one of values.

Input	Keys = ['name', 'age', 'city'] Values = ['Amit', 21, 'Kolhapur']
Expected Output	{'name': 'Amit', 'age': 21, 'city': 'Kolhapur'}

11. Write a Python program to sort a dictionary by keys.

Input	{"c": 3, "a": 1, "b": 2}
Expected Output	{'a': 1, 'b': 2, 'c': 3}

12. Write a Python program to sort a dictionary by values.

Input	{"a": 30, "b": 10, "c": 20}
Expected Output	{'b': 10, 'c': 20, 'a': 30}

13. Write a Python program to find the key with the maximum value in a dictionary.

Input	{"A": 75, "B": 92, "C": 88}
Expected Output	Key with maximum value = B

14. Write a Python program to convert a dictionary into a list of tuples.

Input	{"a": 1, "b": 2, "c": 3}
Expected Output	[('a', 1), ('b', 2), ('c', 3)]

15. Write a Python program to create a nested dictionary for student details.

Input	Student 101: Name = Amit, Marks = 85 Student 102: Name = Neha, Marks = 90
Expected Output	{101: {'name': 'Amit', 'marks': 85}, 102: {'name': 'Neha', 'marks': 90}}

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16. Write a Python program to count the frequency of words in a sentence using a dictionary.

Input	python is easy and python is powerful
Expected Output	{'python': 2, 'is': 2, 'easy': 1, 'and': 1, 'powerful': 1}

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to invert a dictionary and group keys that share the same value.

Input	{'A': 1, 'B': 2, 'C': 1}
Expected Output	{1: ['A', 'C'], 2: ['B']}

2. Write a Python program to merge two dictionaries and add values of common keys.

Input	D1 = {'a': 10, 'b': 20} D2 = {'b': 5, 'c': 15}
Expected Output	{'a': 10, 'b': 25, 'c': 15}

3. Write a Python program to filter dictionary items whose values are greater than or equal to a given threshold.

Input	Marks = {'Amit': 75, 'Neha': 91, 'Ravi': 84} Threshold = 80
Expected Output	{'Neha': 91, 'Ravi': 84}

4. Write a Python program to group words by their first character using a dictionary.

Input	['apple', 'ant', 'ball', 'bat', 'cat']
Expected Output	{ 'a': ['apple', 'ant'], 'b': ['ball', 'bat'], 'c': ['cat']}

5. Write a Python program to calculate the average marks of each student from a nested dictionary.

Input	{ 'Amit': { 'Python': 80, 'ML': 90 }, 'Neha': { 'Python': 92, 'ML': 88 } }
Expected Output	Amit = 85.0 Neha = 90.0

6. Write a Python program to swap keys and values when all dictionary values are unique.

Input	{ 'a': 10, 'b': 20, 'c': 30 }
Expected Output	{ 10: 'a', 20: 'b', 30: 'c' }

7. Write a Python program to find common keys whose values are different in two dictionaries.

Input	D1 = { 'a': 1, 'b': 2, 'c': 3 } D2 = { 'b': 20, 'c': 3, 'd': 4 }
Expected Output	Different common keys = { 'b' }

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8. Write a Python program to rename selected keys in a dictionary using a key-mapping dictionary.

Input	Data = {'fname': 'Amit', 'lname': 'Patil'} Map = {'fname': 'first_name', 'lname': 'last_name'}
Expected Output	{'first_name': 'Amit', 'last_name': 'Patil'}

9. Write a Python program to display character frequencies in descending order of frequency.

Input	mississippi
Expected Output	[('i', 4), ('s', 4), ('p', 2), ('m', 1)]

10. Write a Python program to create a dictionary of numbers and their squares from 1 to n.

Input	n = 5
Expected Output	{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to group a list of words into anagram groups using a dictionary.

Input	['eat', 'tea', 'tan', 'ate', 'nat', 'bat']
Expected Output	{'aet': ['eat', 'tea', 'ate'], 'ant': ['tan', 'nat'], 'abt': ['bat']}

2. Write a Python program to create a word-position index that stores all positions where each word occurs.

Input	python is easy and python is powerful
Expected Output	{'python': [0, 4], 'is': [1, 5], 'easy': [2], 'and': [3], 'powerful': [6]}

3. Write a Python program to flatten a nested dictionary using dot-separated keys.

Input	{ 'student': { 'name': 'Amit', 'marks': { 'Python': 90 } } }
Expected Output	{ 'student.name': 'Amit', 'student.marks.Python': 90 }

4. Write a Python program to recursively merge two nested dictionaries.

Input	D1 = { 'a': 1, 'b': { 'x': 10 } } D2 = { 'b': { 'y': 20 }, 'c': 3 }
Expected Output	{ 'a': 1, 'b': { 'x': 10, 'y': 20 }, 'c': 3 }

5. Write a Python program to update inventory quantities from a list of stock transactions.

Input	Inventory = { 'Pen': 10, 'Book': 5 } Transactions = [('Pen', -3), ('Book', 4), ('Pencil', 6)]
Expected Output	{ 'Pen': 7, 'Book': 9, 'Pencil': 6 }

6. Write a Python program to create a leaderboard with ranks from a dictionary of player scores.

Input	{ 'A': 95, 'B': 82, 'C': 95, 'D': 70 }
Expected Output	Rank 1: A, C (95) Rank 2: B (82) Rank 3: D (70)

7. Write a Python program to create a reverse dependency map from a dictionary of task prerequisites.

Input	{ 'Deploy': ['Test'], 'Test': ['Build'], 'Build': ['Code'] }
Expected Output	{ 'Test': ['Deploy'], 'Build': ['Test'], 'Code': ['Build'] }

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8. Write a Python program to find the highest-scoring subject for each student in a nested dictionary.

Input	{'Amit': {'Python': 90, 'ML': 85}, 'Neha': {'Python': 88, 'ML': 94}}
Expected Output	Amit -> Python (90) Neha -> ML (94)

9. Write a Python program to add two sparse matrices stored as dictionaries with (row, column) keys.

Input	A = {(0,0): 2, (1,1): 3} B = {(0,0): 5, (0,1): 4}
Expected Output	{(0,0): 7, (0,1): 4, (1,1): 3}

10. Write a Python program to simulate an LRU cache of fixed capacity using dictionary insertion order.

Input	Capacity = 3 Requests = A, B, C, A, D
Expected Output	Final cache order (oldest to newest) = [C, A, D]